

Machine execution

Jinyang Li

Lesson Plan: last time

- Basic h/w execution model:
 - CPU fetch the next instruction from memory according to PC
 - Decode and execute instruction (e.g. mov instruction)
 - CPU updates PC to point to next instruction
- ISA (instruction set architecture): x86, ARM, RISC-V
- X86 ISA
 - %rip (PC), 16 general purpose registers
 - mov instruction

Lesson Plan: today

- mov
 - more memory addressing modes
- lea
- arithmetic instructions
- How CPUs realize non-linear control flow

mov: limitation of direct addressing

Direct addressing

- The address must be calculated and stored in the register before each memory access.

```
long a[3] = {1, 2, 3};  
a[2] = 0;
```



1. Calculate address of a[2]
2. Store it in some register, e.g. %rax
3. Do memory write, e.g. movq \$0, (%rax)

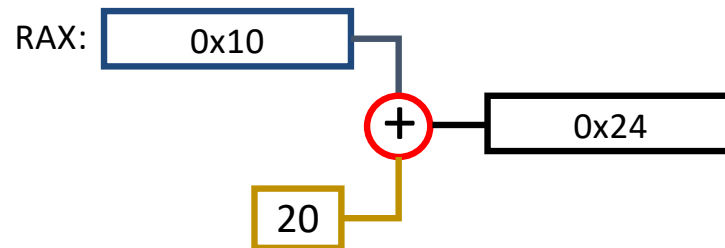
$\&a[2] = \text{address of } a[0] + \text{stride} * 2$

Address mode with displacement

$D(\text{Register}): \text{val}(\text{Register}) + D$

- Register specifies the start of the memory region
- Constant D specifies the offset

Memory Operand: 20(RAX)



Address mode with displacement

$D(\text{Register}): \text{val}(\text{Register}) + D$

- Register specifies the start of the memory region
- Constant D specifies the offset

```
long a[3] = {1, 2, 3};  
a[2] = 0;
```



```
// Suppose register %rax contains address of a[0]  
mov $0, 16(%rax)
```

```
for (i = 0; i < 3; i++)  
    a[i] = 1;
```



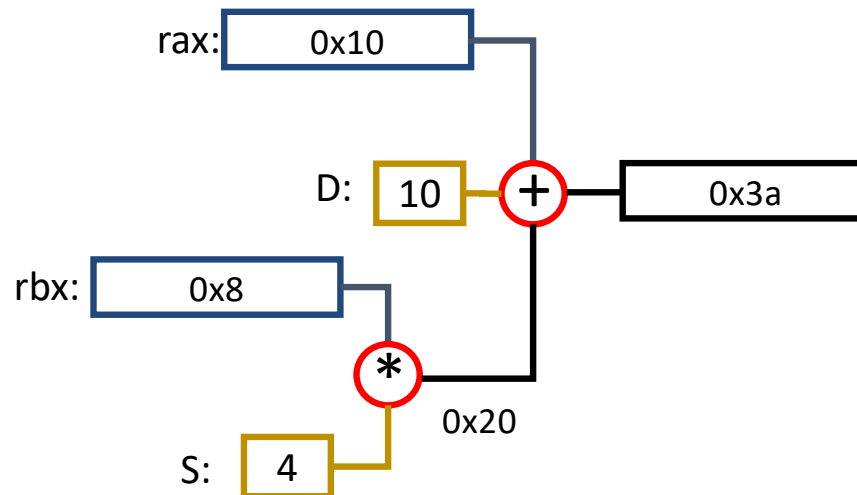
?

X86's Complete Memory Addressing Mode

$D(Rb, Ri, S): val(Rb) + S * val(Ri) + D$

- Rb: Base register
- D: Constant “displacement”
- Ri: Index register (not **%rsp**)
- S: Scale: 1, 2, 4, or 8

Memory operand: 10(%rax, %rbx, 4)



Complete Memory Addressing Mode

$$D(Rb, Ri, S): \text{val}(Rb) + S * \text{val}(Ri) + D$$

If S is 1 or D is 0, they can be omitted

- $(Rb, Ri): \text{val}(Rb) + \text{val}(Ri)$
- $D(Rb, Ri): \text{val}(Rb) + \text{val}(Ri) + D$
- $(Rb, Ri, S): \text{val}(Rb) + S * \text{val}(Ri)$

Address Computation Examples

%rdx	0xf000
%rcx	0x100

Expression	Address Computation	Address
0x8 (%rdx)		
(%rdx,%rcx)		
(%rdx,%rcx,4)		
0x80(,%rdx,2)		

Address Computation Examples

<code>%rdx</code>	<code>0xf000</code>
<code>%rcx</code>	<code>0x100</code>

Expression	Address Computation	Address
<code>0x8(%rdx)</code>	<code>0xf000 + 0x8</code>	<code>0xf008</code>
<code>(%rdx,%rcx)</code>		
<code>(%rdx,%rcx,4)</code>		
<code>0x80(,%rdx,2)</code>		

Address Computation Examples

<code>%rdx</code>	<code>0xf000</code>
<code>%rcx</code>	<code>0x100</code>

Expression	Address Computation	Address
<code>0x8(%rdx)</code>	<code>0xf000 + 0x8</code>	<code>0xf008</code>
<code>(%rdx,%rcx)</code>	<code>0xf000 + 0x100</code>	<code>0xf100</code>
<code>(%rdx,%rcx,4)</code>		
<code>0x80(,%rdx,2)</code>		

Address Computation Examples

<code>%rdx</code>	<code>0xf000</code>
<code>%rcx</code>	<code>0x100</code>

Expression	Address Computation	Address
<code>0x8(%rdx)</code>	<code>0xf000 + 0x8</code>	<code>0xf008</code>
<code>(%rdx,%rcx)</code>	<code>0xf000 + 0x100</code>	<code>0xf100</code>
<code>(%rdx,%rcx,4)</code>	<code>0xf000 + 4*0x100</code>	<code>0xf400</code>
<code>0x80(,%rdx,2)</code>		

Address Computation Examples

<code>%rdx</code>	<code>0xf000</code>
<code>%rcx</code>	<code>0x100</code>

Expression	Address Computation	Address
<code>0x8(%rdx)</code>	<code>0xf000 + 0x8</code>	<code>0xf008</code>
<code>(%rdx,%rcx)</code>	<code>0xf000 + 0x100</code>	<code>0xf100</code>
<code>(%rdx,%rcx,4)</code>	<code>0xf000 + 4*0x100</code>	<code>0xf400</code>
<code>0x80(,%rdx,2)</code>	<code>2*0xf000 + 0x80</code>	<code>0x1e080</code>

Complete addressing mode

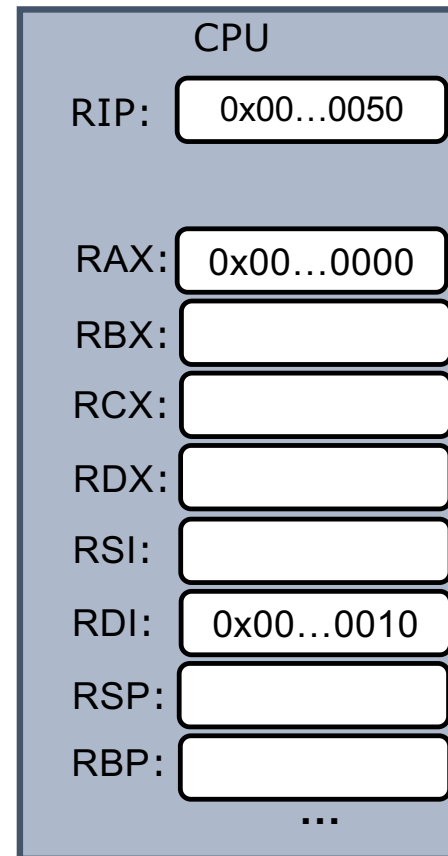
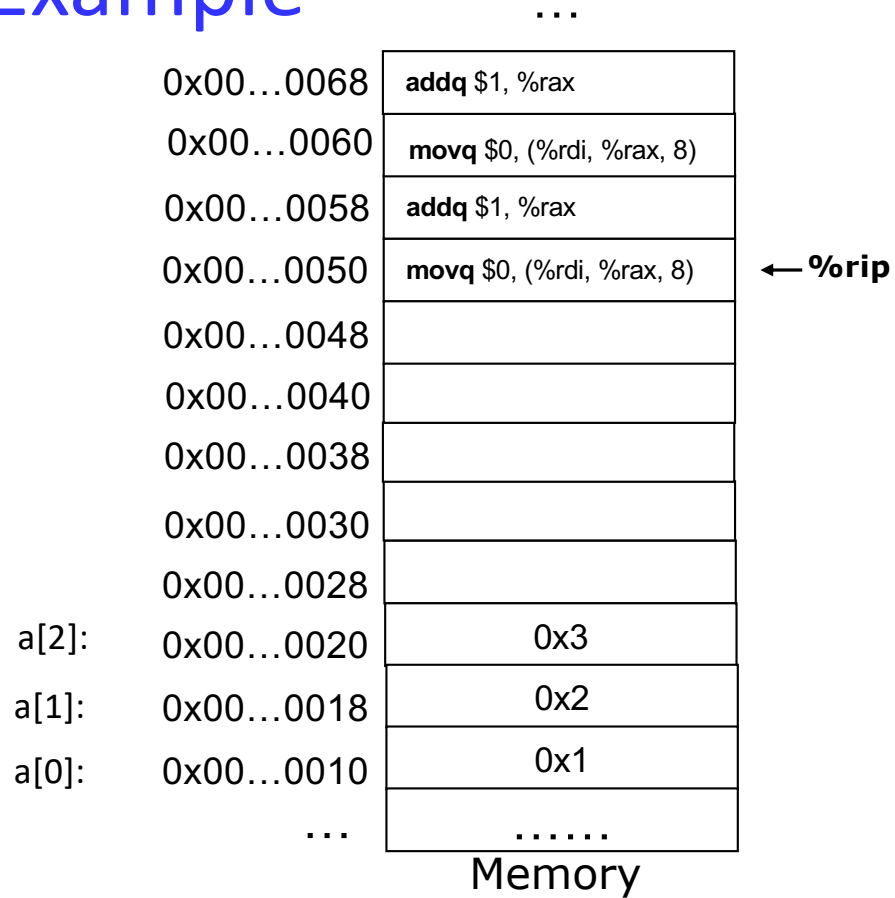
$$D(Rb, Ri, S): \text{val}(Rb) + S * \text{val}(Ri) + D$$

```
long a[] = {1, 2, 3};  
for(int i = 0; i < 3; i++) {  
    a[i] = 0;  
}
```

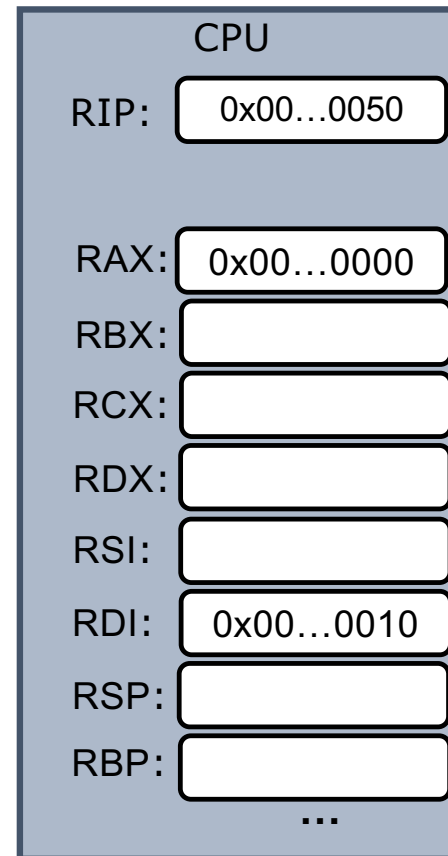
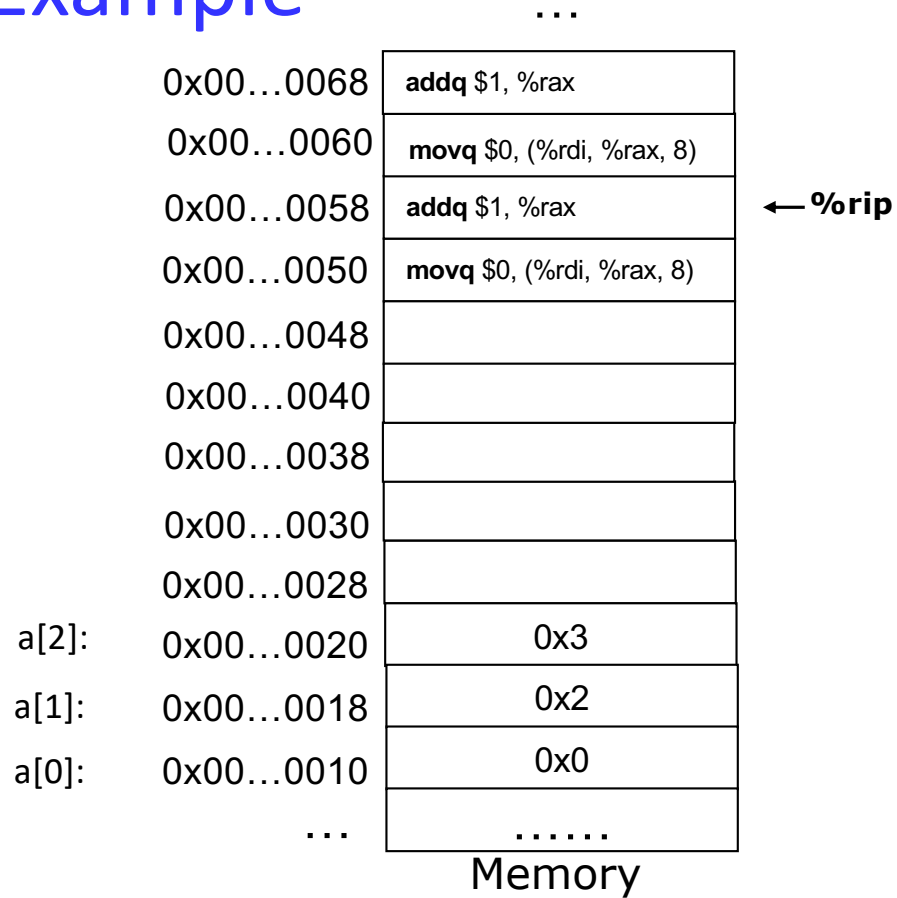


1. Store address of a[0] in some register, say %rdi, store i in another register, say %rax
2. Do memory write: `mov $0, (%rdi, %rax, ?)`

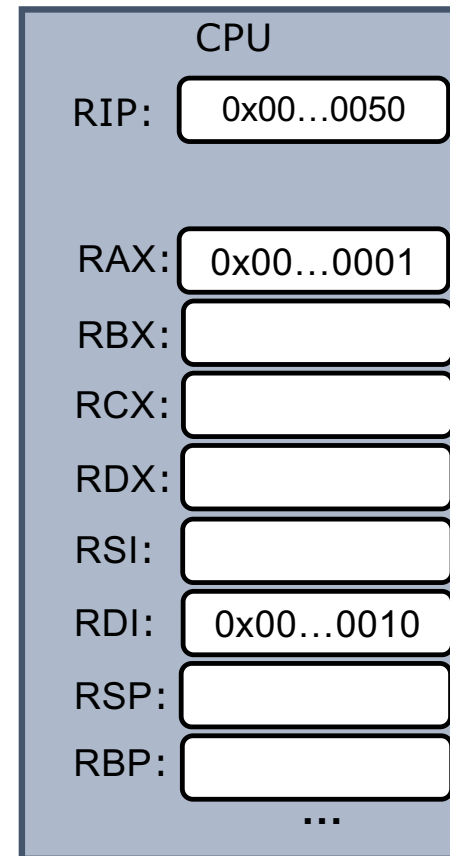
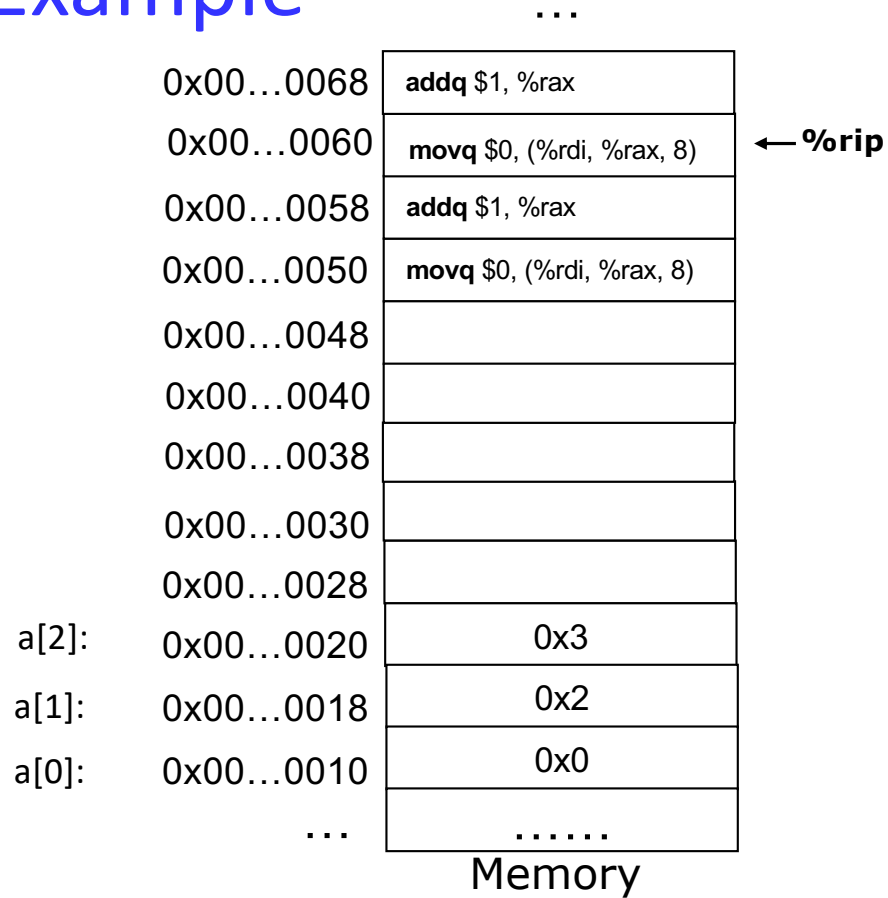
Example



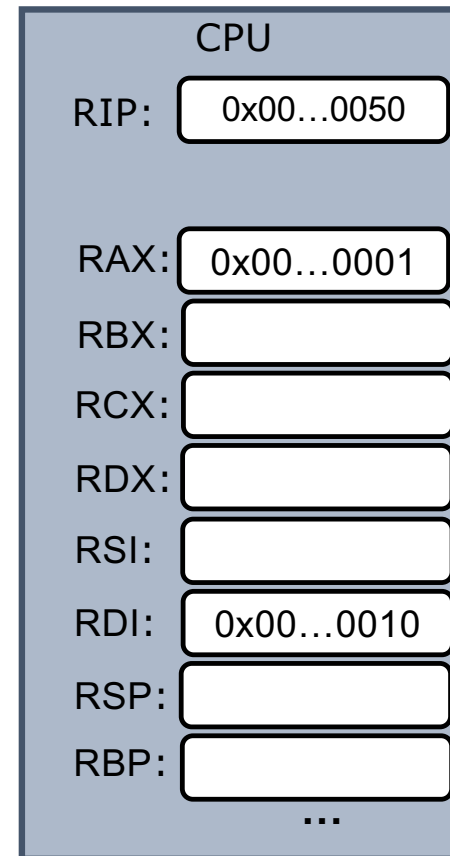
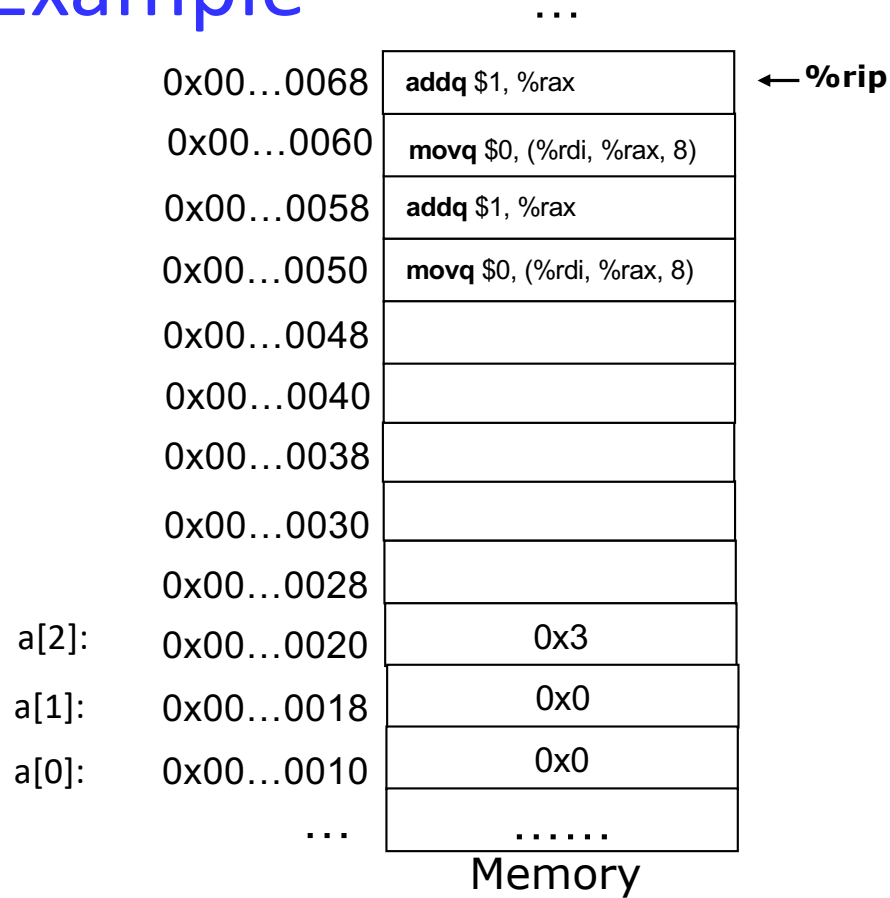
Example



Example



Example

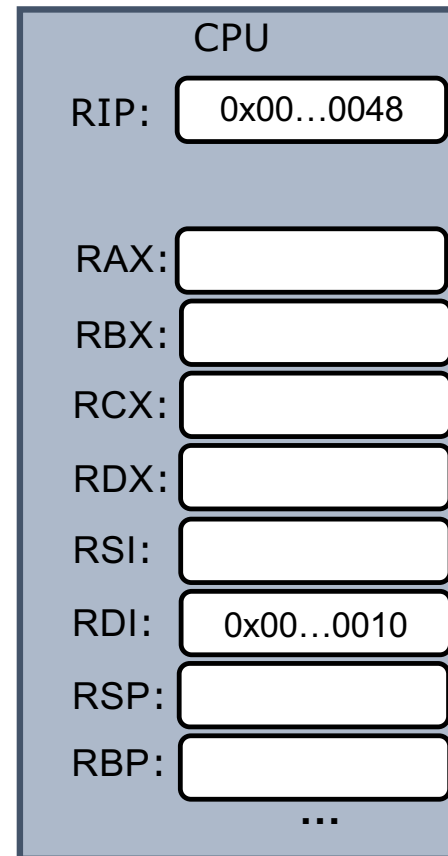
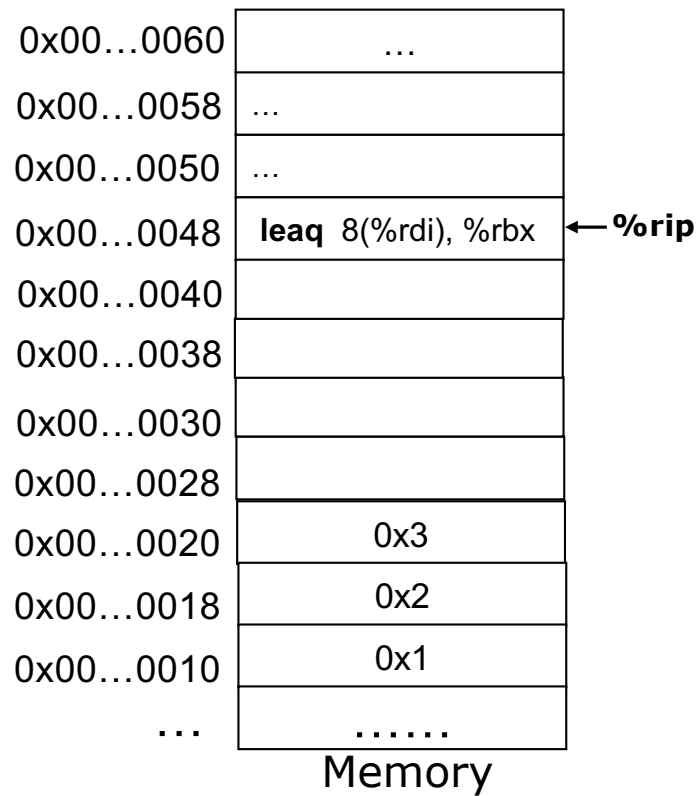


lea instruction

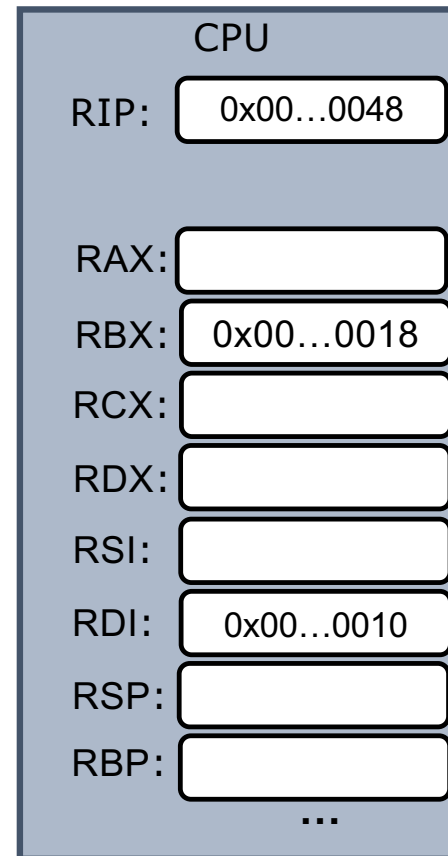
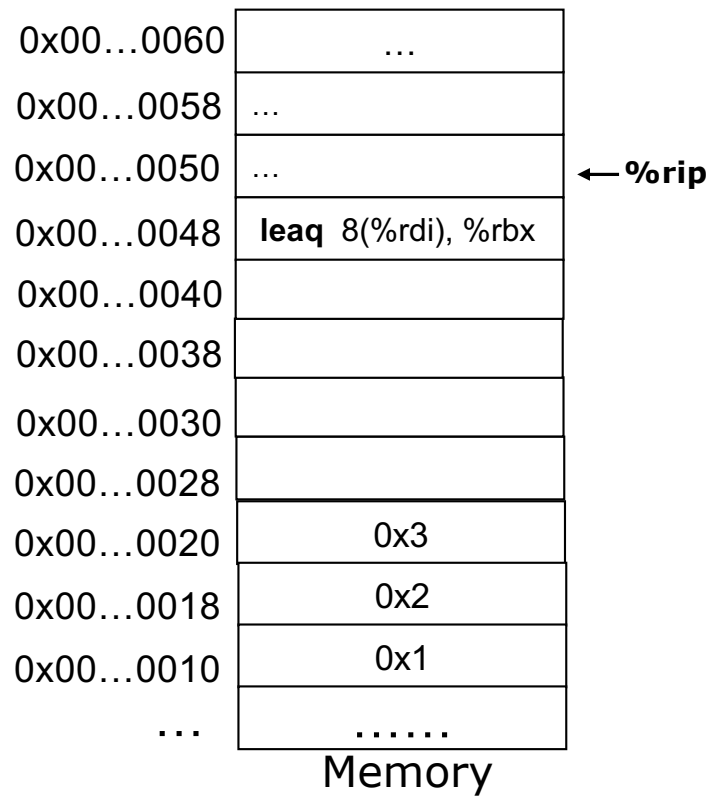
leaq *Source, Dest*

- Short for **L**oad **E**ffective **A**ddress
- Set *Dest* to the address denoted by *Source* address mode expression
- Performs address calculation only; no memory access!

Example



Example



Common use case for lea: calculate address

```
long *p = &a[i];
```



```
leaq (%rdi, %rsi, 8), %rax
```

Another common use case for lea

Lea is used to compute certain simple arithmetic expression

```
long m3(long x)
{
    return x*3;
}
```



```
leaq (%rdi, %rdi,2), %rax
```

Assume %rdi has the value of x

Arithmetic Expression Puzzle

Suppose %rdi, %rsi, %rax corresponds to variables x, y, s, respectively

```
leaq    (%rdi,%rsi,2), %rax  
leaq    (%rax,%rax,4), %rax
```



```
long f(long x, long y)  
{  
    long s = ??;  
    return s;  
}
```

Arithmetic Expression Puzzle

Suppose %rdi, %rsi, %rax contains variable x, y, s respectively

```
leaq    (%rdi,%rsi,2), %rax  
leaq    (%rax,%rax,4), %rax
```



```
long f(long x, long y)  
{  
    long s = 5(x + 2y);  
    return s;  
}
```

Basic Arithmetic Operations

addq	Src, Dest	$\text{Dest} = \text{Dest} + \text{Src}$
subq	Src, Dest	$\text{Dest} = \text{Dest} - \text{Src}$
imulq	Src, Dest	$\text{Dest} = \text{Dest} * \text{Src}$

incq	Dest	$\text{Dest} = \text{Dest} + 1$
decq	Dest	$\text{Dest} = \text{Dest} - 1$
negq	Dest	$\text{Dest} = -\text{Dest}$

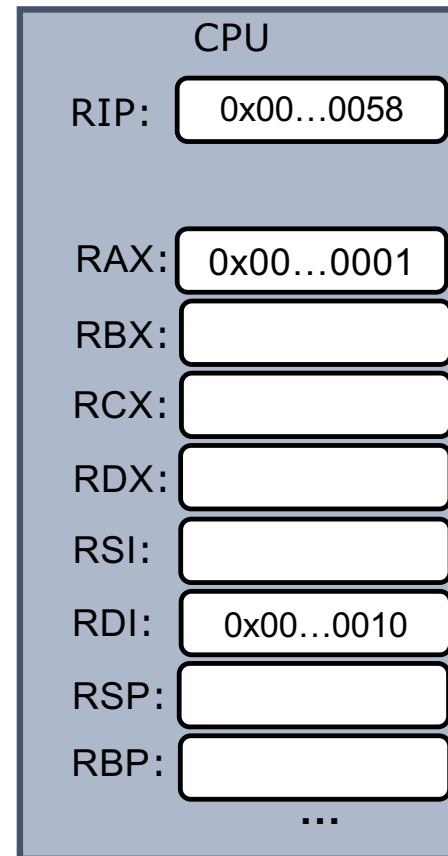
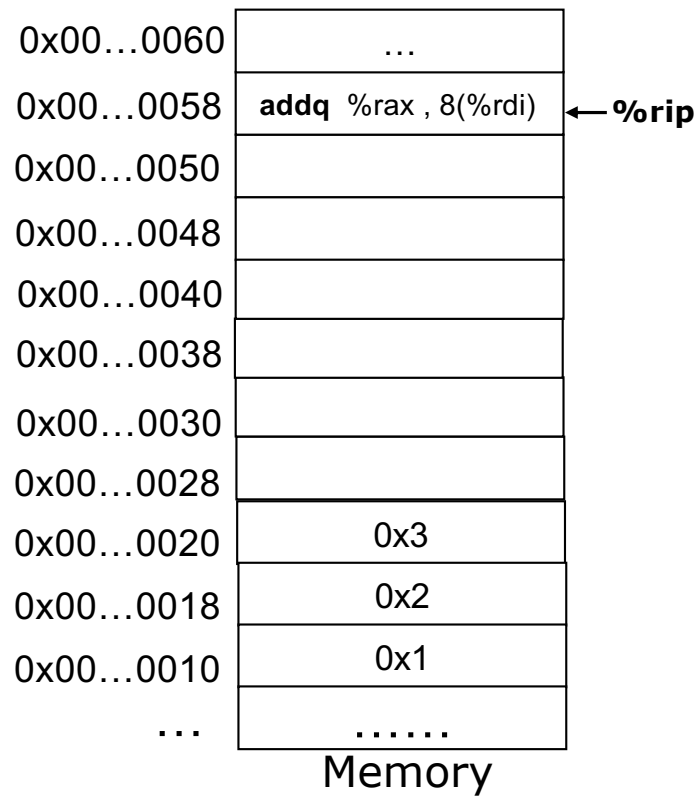
Bitwise Operations

salq	Src, Dest	Dest = Dest << Src
sarq	Src, Dest	Dest = Dest >> Src
shlq	Src, Dest	Dest = Dest << Src
shrq	Src, Dest	Dest = Dest >> Src
xorq	Src, Dest	Dest = Dest ^ Src
andq	Src, Dest	Dest = Dest & Src
orq	Src, Dest	Dest = Dest Src
notq	Dest	Dest = ~Dest

Arithmetic left shift
Arithmetic right shift
Logical left shift
Logical right shift



Example



Example

0x00...0060	...	← %rip
0x00...0058	addq %rax , 8(%rdi)	
0x00...0050		
0x00...0048		
0x00...0040		
0x00...0038		
0x00...0030		
0x00...0028		
0x00...0020	0x3	
0x00...0018	0x3	
0x00...0010	0x1	
...		

Memory

CPU	
RIP:	0x00...0058
RAX:	0x00...0001
RBX:	
RCX:	
RDX:	
RSI:	
RDI:	0x00...0010
RSP:	
RBP:	
...	

Lesson Plan: today

- mov
 - complete memory addressing
- lea
- arithmetic instructions
- How CPUs realize non-linear control flow

How is control flow realized?

if ... else  ???

for loop

while loop

...

Control flow uses RFLAGS register

PC: Program counter

- Store memory address of next instruction
- Also called “RIP” in x86_64

IR: instruction register

- Store the fetched instruction

General purpose registers:

- Store operands and pointers used by program

Program status and control register:

- Contain status of the instruction executed
- All called “**RFLAGS**” in x86_64

How control flow uses RFLAGS register

- RFLAGS is a special purpose register
- Different bits represent different status flags
- Certain instructions set status flags
 - Regular arithmetic instructions
 - Special flag-setting instructions: **cmp**, **test**, **set**
- **jmp** instructions use flags to determine value of %rip

EFLAGS register: ZF

- ZF (Zero Flag):
 - Set if the result of the instruction is zero; cleared otherwise.

```
movq $2, %rax  
subq $2, %rax
```

EFLAGS register: SF

- SF (Sign Flag):
 - Set to be the most-significant bit of the result.

```
movq $2, %rax  
subq $10, %rax
```

EFLAGS register: CF

- CF (Carry Flag):
 - Set if adding (or subtracting) two numbers carries out (or borrow out) of msb

➡ i.e. Set if overflow for unsigned integer arithmetic

```
movq $0xffffffffffffffff, %rax  
addq $2, %rax
```

```
movq $0, %rax  
subq $1, %rax
```

EFLAGS register: OF

- OF (Overflow Flag):
 - Set if there is carry-in but no carry-out of MSB
 - or, there is no carry-in but there's carry-out of MSB



Set if overflow for signed integer (2's complement) arithmetic.

```
movq $0x7fffffffffffffffff, %rax  
addq $1, %rax
```

```
movq $0x8000000000000000, %rax  
addq $0xffffffffffffffff, %rax
```

CF and OF are different flags

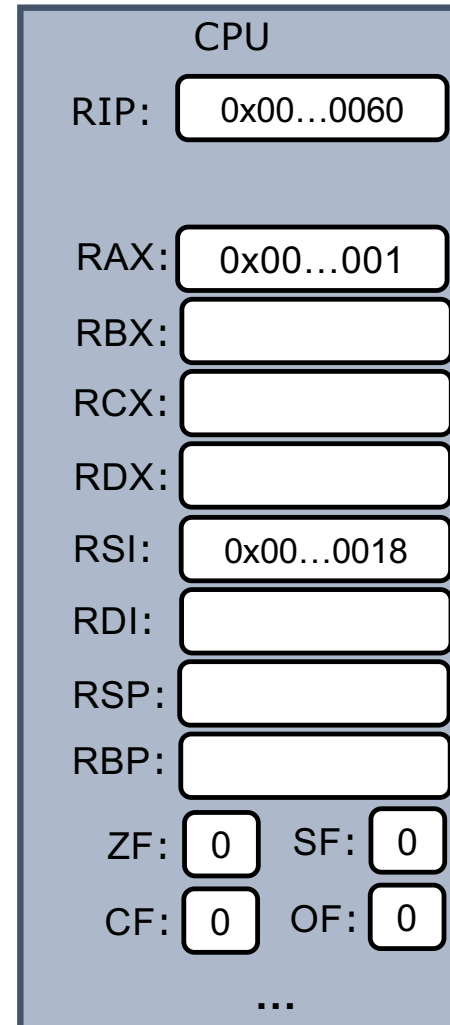
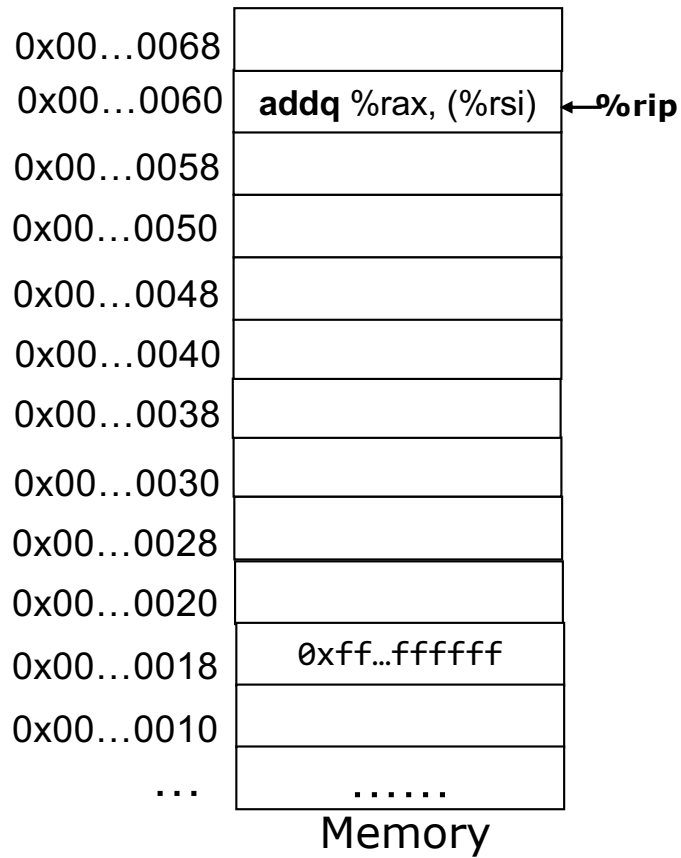
- CPU does not care if data represents signed or unsigned integer:
 - Same underlying arithmetic hardware circuitry.
 - OF and CF flags are set by examining various carry bits
- Up to programmer/compiler to use either CF or OF flag

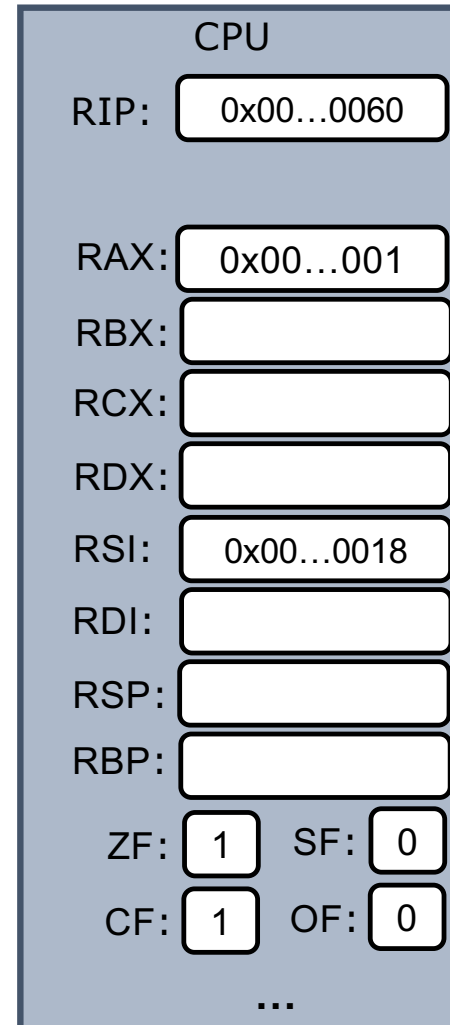
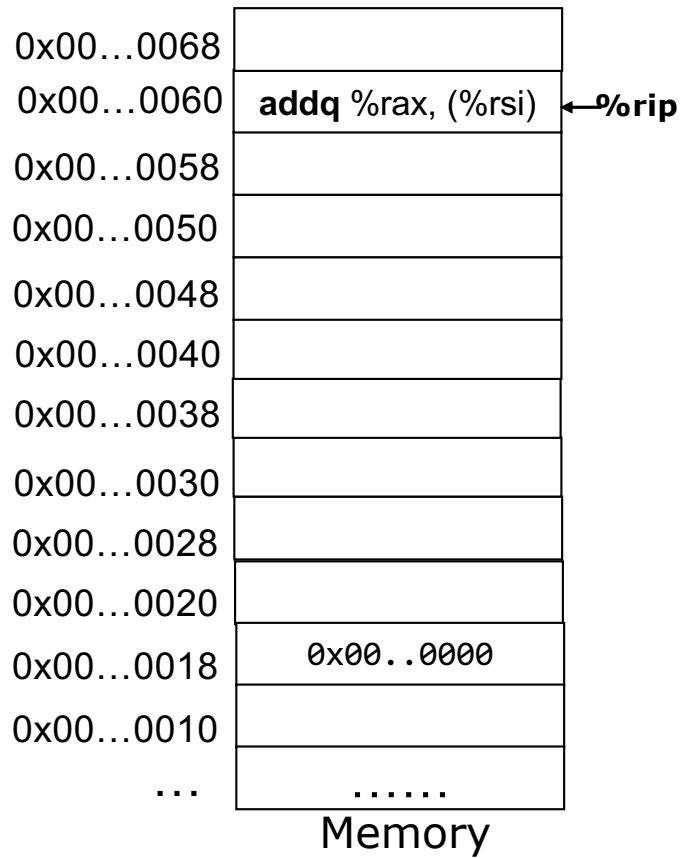
Status flags summary

flag	status
ZF (Zero Flag)	set if the result is zero.
SF (Sign Flag)	set if the result is negative.
CF (Carry Flag)	Overflow for unsigned-integer arithmetic
OF (Overflow Flag)	Overflow for signed-integer arithmetic

Set by arithmetic instructions, e.g. add, inc, and, sal

Not set by **lea**, **mov**





Summary

- X86 ISA
 - %rip, 16 general purpose registers
 - mov
 - Lea
 - Various arithmetic instructions
 - RFLAGS: ZF, SF, CF, OF